CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Carbon compounds have low melting and boiling points. Reason (R): Intermolecular forces in covalent compounds are weak. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The balanced equation for the complete combustion of methane is: 1

(a) $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ **(b)** $2\,CH_4 + O_2 \longrightarrow 2\,CO_2 + H_2O$ **(c)** $CH_4 + O_2 \longrightarrow CO_2 + H_2O$ **(d)** $CH_4 + 3\,O_2 \longrightarrow CO_2 + 2\,H_2O$

3. A hydrocarbon X decolourises bromine water and has a molecular mass of 28 u. X is: 1

(a) ethyne **(b)** methane **(c)** ethene **(d)** ethane

4. Assertion (A): Unsaturated hydrocarbons burn with a sooty flame. Reason (R): They have a higher percentage of carbon so burn incompletely. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. In the balanced equation $2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$, the coefficient of water is: 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 7

6. An organic compound Y gives effervescence with $NaHCO_3$ and is used as a preservative. Y is: 1

(a) ethanoic acid **(b)** methanal **(c)** ethanol **(d)** ethane

7. Assertion (A): Diamond is a poor conductor of electricity but graphite conducts. Reason (R): In graphite, each carbon is bonded to four others. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Natural gas burns with a clean blue flame because its main constituent, methane, is: 1

(a) an ester **(b)** an alcohol
(c) an unsaturated hydrocarbon **(d)** a saturated hydrocarbon

9. **Assertion (A):** Carbon forms a very large number of compounds. 1
Reason (R): Carbon is tetravalent and shows the property of catenation.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Ethanol is oxidised to ethanoic acid by alkaline potassium permanganate. 1

Reason (R): Acidified potassium dichromate is also an oxidising agent that converts ethanol to ethanoic acid.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the balanced equation for the complete combustion of ethane and name the products. 2

12. An alkane has 22 covalent bonds. Find its molecular formula. 2

13. Why can propyne add hydrogen? How many hydrogen molecules add to one molecule of propyne? 2

14. Calculate the percentage of carbon in ethane. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the structural formula of propene and state two ways in which it differs from propane. 3

16. Determine the number of C-C and C-H bonds in butane and in isobutane and comment. 3

17. State the effect of chain length on physical state in the alkane series with examples. 3

OR

Write the balanced equation for the complete combustion of methane and state the nature of the flame. 3

18. Explain why graphite conducts but diamond does not, and why fullerene is a molecular solid. 3

OR

Compute the molecular mass of the 5th member of the alkane series and the difference between members. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Alkanes form a homologous series with the general formula C_nH_{2n+2}. The first member is methane, the second is ethane and the third is propane. The fourth member is butane and the fifth is pentane. 5

(i) Write the molecular formula of the alkane with five carbon atoms.

(ii) Name the first member of the alkyne series.

(iii) What is the difference in molecular mass between two consecutive members of a homologous series?

(iv) Find the number of covalent bonds in propane and explain why propane is called a saturated compound.

20. Write the equation for the complete combustion of ethanol and explain why ethanol can be used as a fuel additive. 5

OR

Explain the allotropy of carbon with bonding diagrams (described), property and use of diamond, graphite, fullerene and why they burn to give the same product. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND